

Unit 13: Samba Servers

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Objectives

After studying this unit, you will be able to

- Understand the SAMBA server
- Installing, starting and stopping the server
- SAMBA configuration with SWAT, starting SWAT service
- Adding SAMBA user
- Creating and configuring SAMBA share

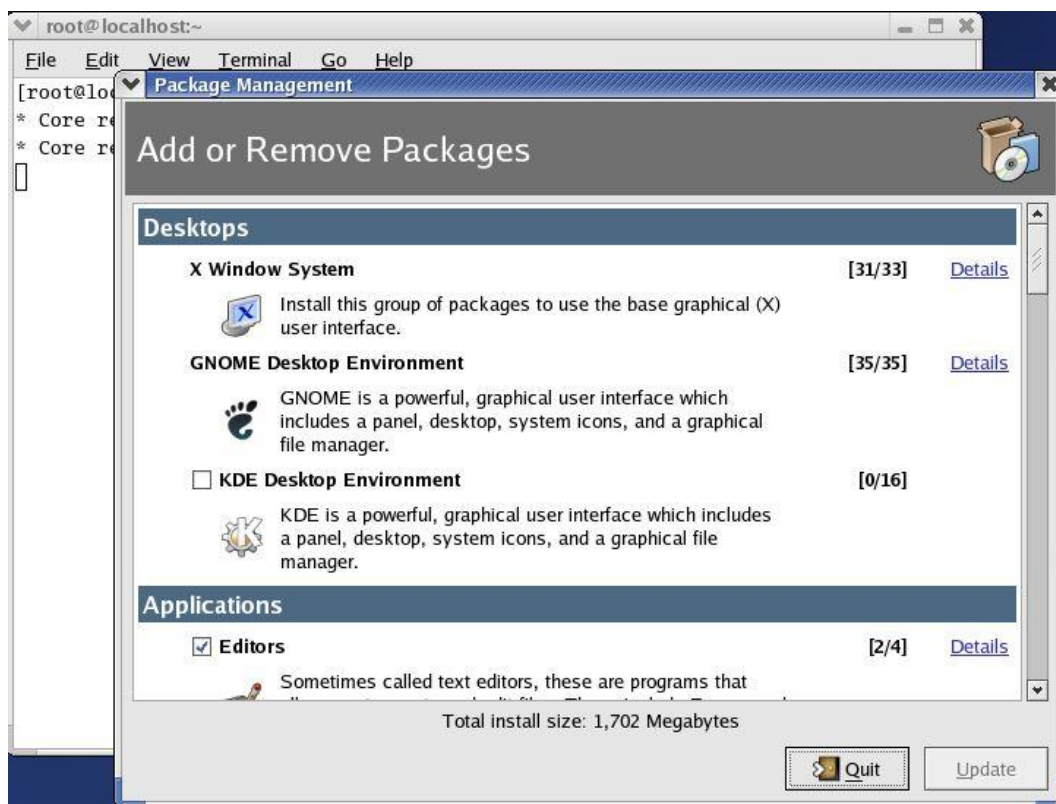
Introduction

Samba is an implementation of the Windows SMB and CIFS protocols on Unix. The Samba project was started by Andrew Tridgell, who wanted to mount his Unix server disk space onto a DOS PC. When he'd solved the problem, Tridgell discovered that what he'd built was an implementation of the SMB (servermessage block) protocol – a protocol for sharing files and other resources. Tridgell named his implementation

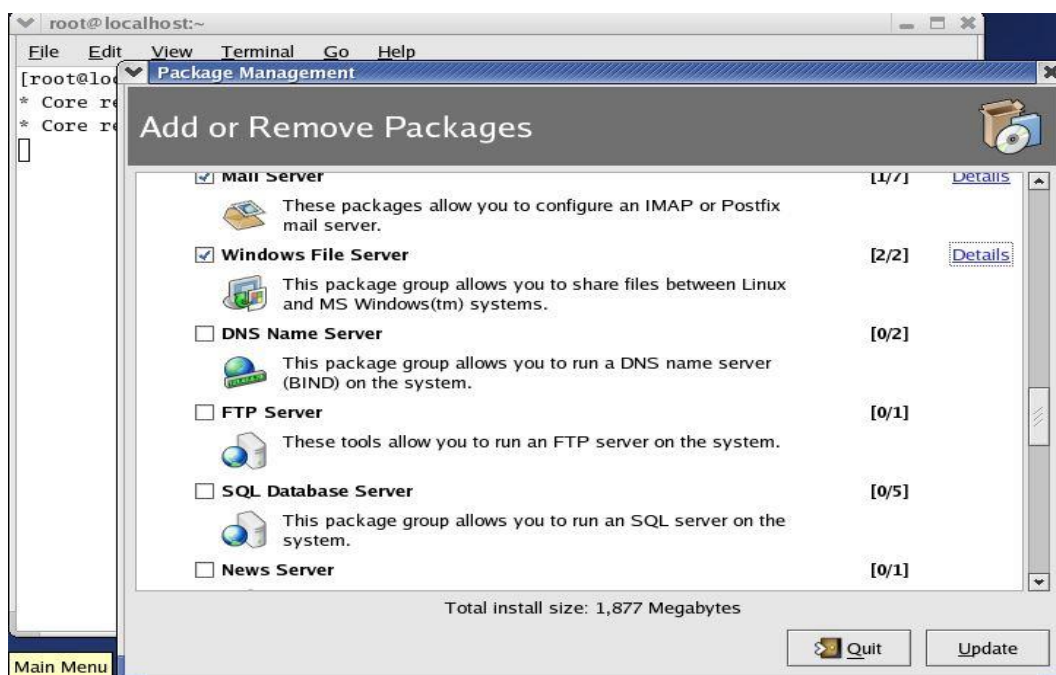
Samba, and published version 1.0 in early 1992. Since that time, the Samba project has grown tremendously, and today there is still Samba development going on in the open-source community. So, Samba is a collection of programs that make it possible to share files and printers between computersequipped to use the SMB protocol – Windows by default, Linux/Unix with Samba, and (more recently) MacOS X. Samba is freely available under GNU General Public License and is included as part of the Red HatLinux 9 distribution.

13.1 Installing SAMBA

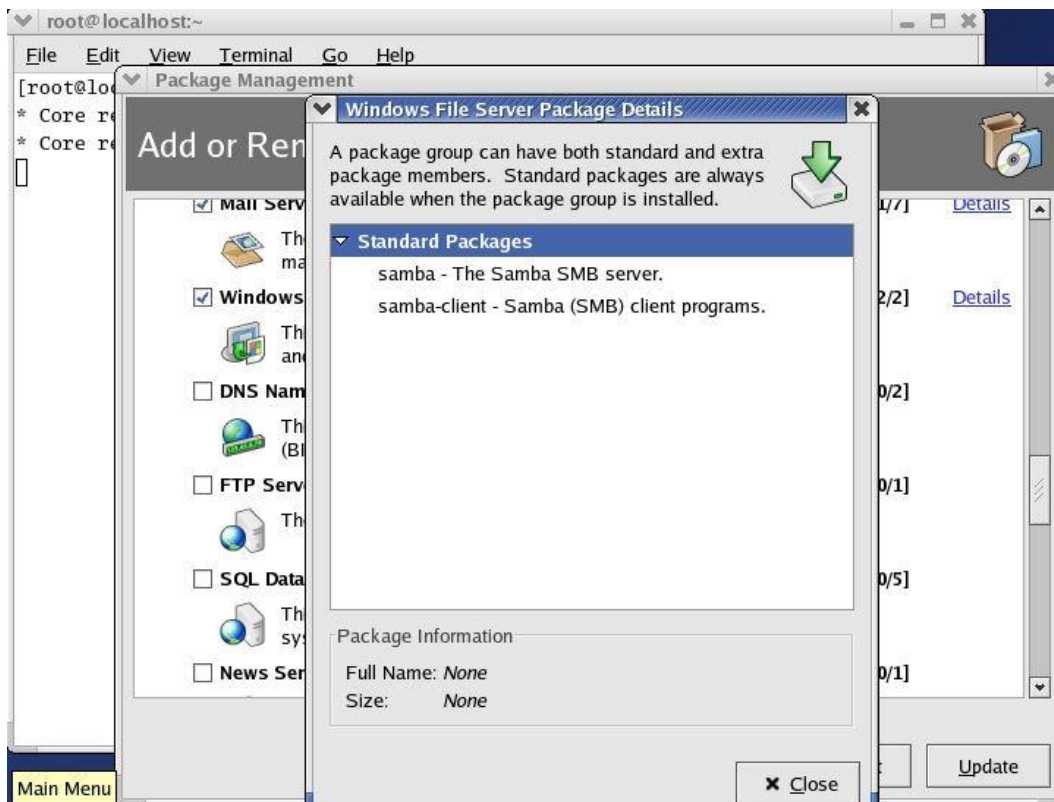
The installation of SAMBA Server suite can be done via the RPM GUI tool. One of the way of this is Main Menu | System Settings | Add/Remove Applications and another way is by using the terminal and writing the command \$ redhat-config-packages.



In package management window, we will look for the servers category. In servers, we will go for Windows File Server.



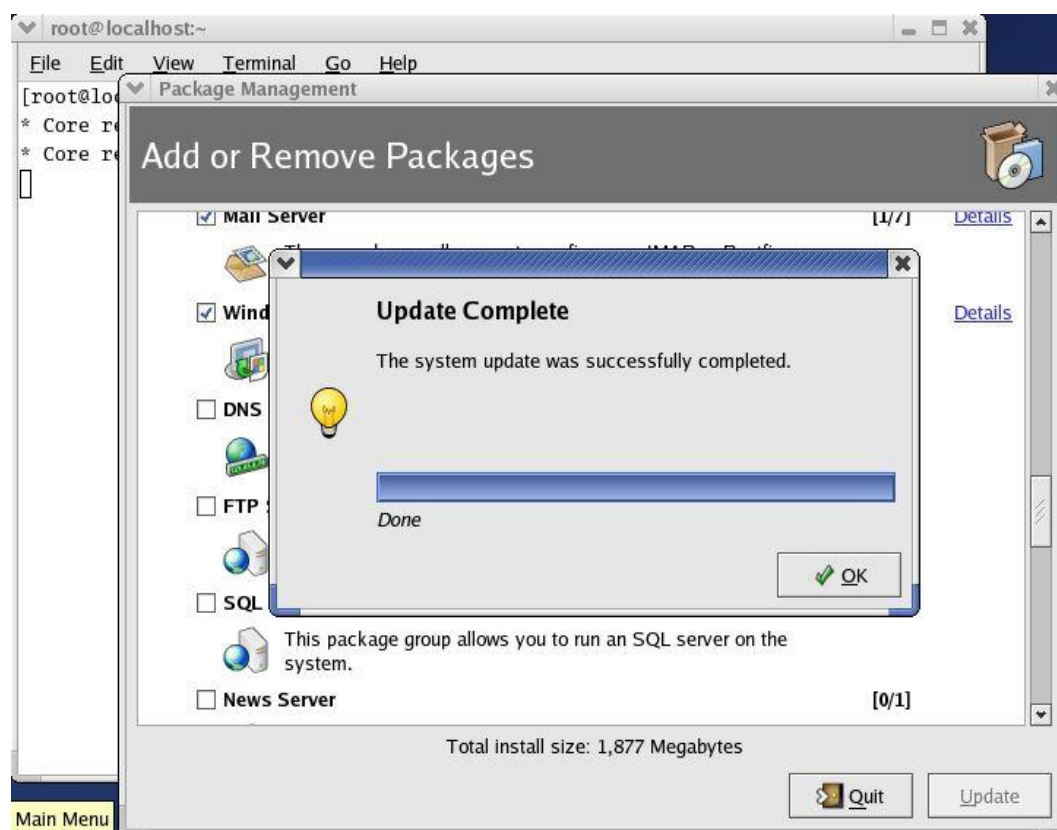
By clicking on the details, we can see the packages available in this. There are two standard packages available in this. These are: samba which is the Samba SMB server and samba-client which is SMB client programs.



After clicking on update button, the packages will be installed. The popup will show the required disk space. After clicking on continue, it will queue the packages for installation.

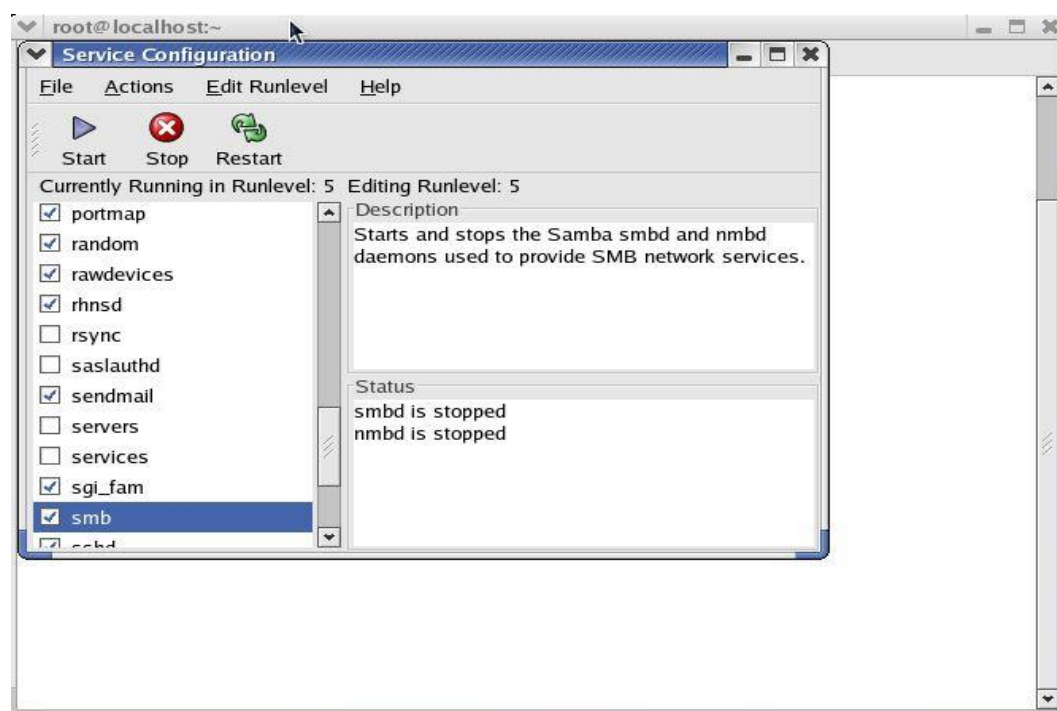


After few seconds, the update will complete. Click on OK.



13.2 Starting and Stopping the Samba Service

There are several ways to start and stop the Samba service. Once again, we can do so via the Service Configuration GUI tool. To launch the tool, select Main Menu | System Settings | Server Settings | Services or type the following command at the command line: `$ redhat-config-services`. Click on smb in this. Now the service is stopped. We need to start it by clicking on the start button.



It's also a good idea to check the checkbox, to configure the samba service to start automatically whenever you boot up the system. For example, if you ever have to perform an emergency reboot on your file server, then the "automatic start" configuration means that the file server is immediately available to users after the reboot. When you've done this, select File | Save Changes to save your new setting.

Alternatively, you can also stop and start smb service at the command line, using the service command to run the /etc/rc.d/init.d/smb script we mentioned earlier. Typing the script name at the command line like this reveals the possible usages:

```
# servicesmb
```

```
Usage: /etc/rc.d/smb {start|stop|restart|reload}status|condrestart}
```

As you can see, it works in much the same way as the httpd and vsftpd scripts we've seen in earlier sections of this chapter. So, to start the service we'd type this:

```
# servicesmb start
```

```
Starting SMB services: [ OK ]
```

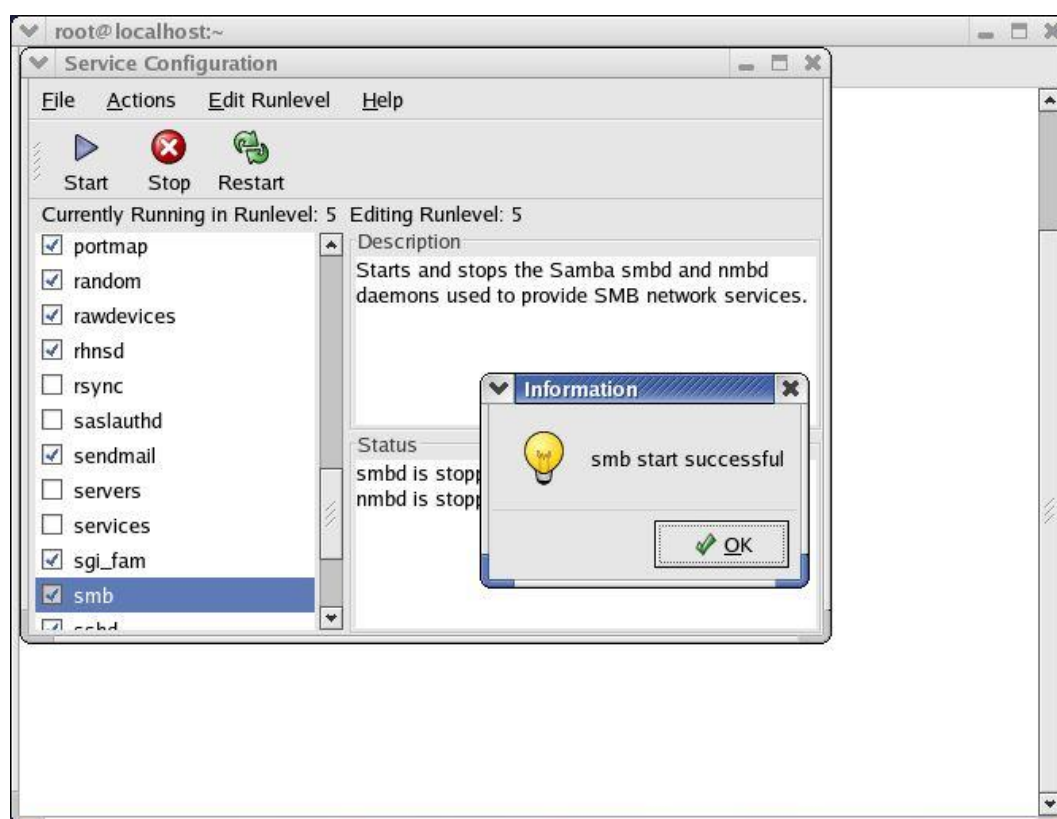
```
Starting NMB services: [ OK ]
```

This command starts both SMB and NMB (NetBIOS name server), which are both services related to Samba. To stop the service, we'd type this:

```
# servicesmb stop
```

```
Shutting down SMB services: [ OK ]
```

```
Shutting down NMB services: [ OK ]
```



13.3 Samba Configuration Files and Utilities

Samba's configuration files are contained in the directory `/etc/samba`.

Configuration File	Description
<code>smb.conf</code>	This is the main configuration file for Samba.
<code>lmhosts</code>	This contains Samba's NetBIOS-to-IP address mappings.
<code>secrets.tdb</code>	This is the Samba secrets database. It stores private information such as the local SID and machine trust password. It is generated by machine and cannot be read in a text editor.
<code>smbusers</code>	This is a text file that maps your Linux system's users to various SMB-specific usernames.
<code>smbpasswd</code>	This is an encrypted password file. This file doesn't exist when you first install Samba, but is created when you add Samba users.

Some of the most important utilities provided by Samba. They're all contained in the directory `/usr/bin`.

Program	Purpose
<code>smbclient</code>	This is an FTP-like client, used to access SMB/CIFS resources on a file server.
<code>smbadduser</code>	This is a script, used for adding Samba users. It updates the <code>smbusers</code> and <code>smbpasswd</code> files.
<code>smbpasswd</code>	This changes a Samba user's SMB password. It is similar to the Unix <code>passwd</code> command.
<code>smbmount</code>	This is used to mount an SMB filesystem.
<code>smbumount</code>	This is used to unmount an SMB file system.
<code>smbstatus</code>	This lists the current Samba connections.
<code>testparm</code>	This checks the <code>smb.conf</code> configuration file for correctness.
<code>nmblookup</code>	This is used to query NetBIOS names and map them to IP addresses in a network using NetBIOS over TCP/IP.

In addition, we'll also make use of the script `/etc/rc.d/init.d/smb`, which we use to start and stop the Samba fileservice.

13.4 Samba Configuration with SWAT

Perhaps the easiest way to configure Samba is by using the Samba Web Administration Tool(SWAT). SWAT is a web-based interface, which means that you can use it to configure and manage your Samba server through a web browser – if you want, you can even do it remotely across a network or even across the Internet.

Installing SWAT

- 1) Launch a terminal window, and switch to the root user account by using this command:
`$ su-`
- 2) Insert Red Hat Linux 9 distribution Disk 2. Change to the directory on the CD that contains the RPM package files: `# cd /mnt/cdrom/RedHat/RPMS`
- 3) Use the `ls` command to find out the exact version of `samba-swat` contained on the disk. For example:
`# ls samba-swat*.rpm`
`samba-swat-2.2.7a-6.i386.rpm`

If this command doesn't find any matches, then remove the disk, replace it with Disk 1 or Disk 2, and return to Step 2.

- 4) Install the `samba-swat` package you've just found, by using the `rpm` command:
`# rpm -ivh samba-swat-2.2.7a-6.i386.rpm`

Starting the SWAT Service

Starting the SWAT service is a two step process:

- 1) Launch `gedit` or your favorite text editor, and open the file `/etc/xinetd.d/swat`. This is the configuration file for the SWAT service, and it looks like this:

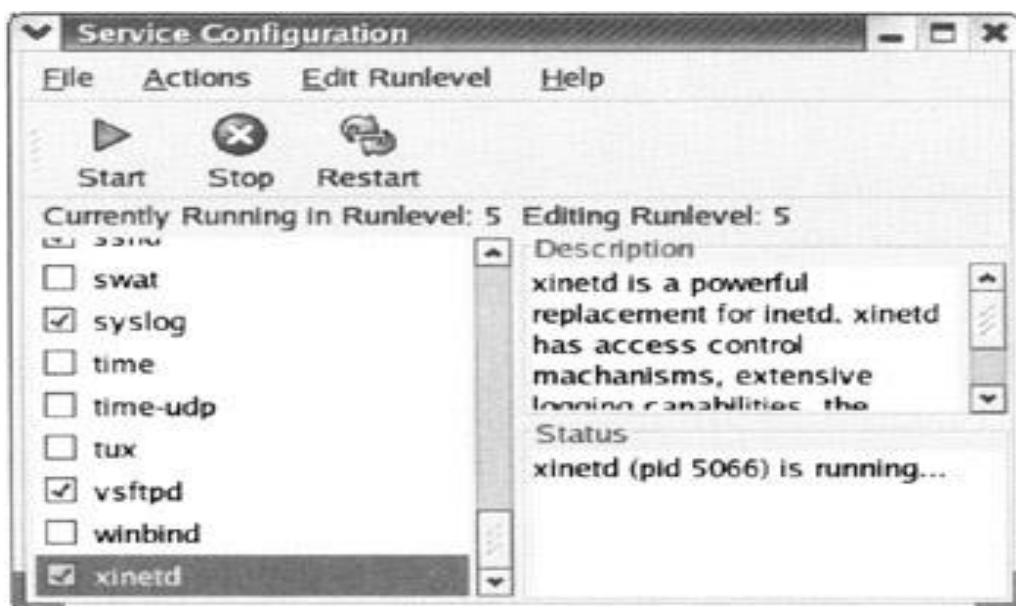
```
# default: off
# description: SWAT is the Samba Web Admin Tool. Use swat \
# to configure your Samba server. To use SWAT, \
# connect to port 901 with your favorite web browser.

service swat
{
    port = 901
    socket_type = stream
    wait = no
    only_from = 127.0.0.1
    user = root
    server = /usr/sbin/swat
    log_on_failure += USERID
    disable = yes
}
```

Change `disable` value, like this: `disable = no`. Save the file, and close your text editor.

2) Restart the xinetd service. (We must do this because SWAT runs as an xinetd service.

To do this, first launch the Service Configuration GUI tool by selecting Main Menu | System Settings | Server Settings | Services. Then locate and select the service called xinetd, and click Restart. You'll get a dialog to confirm that the restart was successful; then you can exit the tool by clicking on Quit.



Using SWAT for the First Time

Now we can test SWAT. Open a browser on the Linux server and type in the URL <http://localhost:901>. You'll be prompted for a username and password – use the system's root username and password. Then you should see the SWAT HOME page:



There are also eight buttons listed across the top of this page, which provide access to SWAT's various utilities:

- **HOME** – the home page of SWAT, and the first page that you see when you fire up the SWAT interface.
- **GLOBALS** – for setting the global variable values of the `/etc/samba/smb.conf` configuration file.
- **SHARES** – for creating and deleting Samba shares, and setting Samba parameters.
- **PRINTERS** – for creating and deleting Samba printer shares and setting printer parameters.
- **WIZARD** – like GLOBALS, this is also for setting various values in `/etc/samba/smb.conf`.
- **STATUS** – for viewing the Samba server's status, and starting and stopping Samba-related services
- **VIEW** – for viewing the content of the `/etc/samba/smb.conf` configuration file
- **PASSWORD** – for adding, removing, enabling, and disabling Samba users, and for setting Samba users' passwords

Adding a Samba User

To grant a system account access to the Samba services, you can use SWAT's PASSWORD feature. Once you've logged into SWAT using the root username and password, click the PASSWORD button, and you'll see the following screen:



In the Server Password Management section, enter the name of an existing account on the system, and supply a password. Then click the Add New User button. This will add an entry to Samba's `smbpasswd` configuration file, to indicate that this user has access to Samba's services.

Creating and Configuring a Samba Share

- 1) Create a directory, which we'll call `/share`, to be used for the file server. With root permission, you can do this at the command line using this command: `# mkdir /share`
- 2) If you haven't done so already, use the Mozilla browser (or your favorite web browser) to browse to the SWAT home page at `http://localhost:901`, and log in using the root user account.
- 3) Click on the GLOBALS toolbar icon. In the Base Options section, use the Workgroup field to enter the name of the workgroup that you want your server to appear in when clients use it. (If you haven't set up a workgroup, then this is probably the default value,

- WORKGROUP). You should also name the service, by entering a value in the server string field – it can be any string that you want your Samba clients to see.
- 4) Now click on SHARES toolbar button at the top of the screen. We'll share our /share directory by giving it a share name; let's call it linuxbox-share. You should first check the entries in the drop-down list on this page, to ensure that your chosen share name hasn't already been used for a different share; then, enter the share name in the Create Share field.
 - 5) Now click on Create Share button to create the share. This will present you with another screen in which you can specify the properties of your share. You will certainly need to set the path field to the path of your share directory (in this example, it's /share).
 - 6) Here, we've set the read only field to No to make it a writeable share; the browseable field to Yes, to allow the contents of the /share directory to be visible; and the available field to Yes, to "enable" the share (that is, make it available to users). We've also added a comment to remind us what the share is for. When you're done, click on the Commit Changes button to commit these settings to the Samba configuration file.
 - 7) Restart the Samba service, so that the configuration changes can take effect. To do this, click on STATUS button and then click on the Restart smbd button to restart the service. Wait for the page to reload, and then click on the Restart nmbd button to restart that service too.

Summary

- Samba is an implementation of the Windows SMB and CIFS protocols on Unix.
- Samba is freely available under GNU General Public License and is included as part of the Red Hat Linux 9 distribution.
- There are two ways to install the SAMBA Server suite via the RPM GUI tool: Main Menu | System Settings | Add/Remove Applications and \$ redhat-config-packages.
- There are two standard packages available in this: samba and samba-client.
- There are two ways to start the SAMBA Service, we can use the Service Configuration tool. These are: Main Menu | System Settings | Server Settings | Services and \$ redhat-config-services.
- If you ever have to perform an emergency reboot on your file server, then the "automatic start" configuration means that the file server is immediately available to users after the reboot.

Keywords

- **Samba:** It is a collection of programs that make it possible to share files and printers between computers equipped to use the SMB protocol – Windows by default and Linux/Unix with Samba.
- **Windows File Server:** This package group allows you to share files between Linux and MS windows system.
- **smb.conf:** This is the main configuration file for Samba.
- **Smbclient:** This is an FTP-like client, used to access SMB/CIFS resources on a file server.
- **SWAT:** It is a web-based interface, which means that you can use it to configure and manage your Samba server through a web browser – if you want, you can even do it remotely across a network or even across the Internet.

- **Xinetd:** xinetd is an open-source super-server daemon which runs on many Unix-like systems and manages Internet-based connectivity.

Self Assessment

1. In package group "Windows File Server", which standard packages are available?
 - A. samba
 - B. samba-client
 - C. Both of the above
 - D. None of the above
2. Which of these is the main configuration file of SAMBA?
 - A. secrets.tdb
 - B. smb.conf
 - C. samba.conf
 - D. imhosts
3. Starting of SWAT service is _____ step process.
 - A. One
 - B. Two
 - C. Three
 - D. Four
4. SAMBA server on UNIX is an implementation of
 - A. Windows SMB
 - B. CIFS protocol
 - C. Both of the above
 - D. None of the above
5. Which server group will be chosen for installing SAMBA server?
 - A. DNS name server
 - B. FTP server
 - C. Mail server
 - D. Windows File server
6. Which of these commands will stop the smb service?
 - A. # servicesmb stop
 - B. # servicesmb end
 - C. # service stop smb
 - D. # service end smb
7. Which of these is unmount an SMB file system?
 - A. smbunmount

- B. smbmount
 - C. smbmount
 - D. None of the above
8. For configuration of SAMBA, which service is required?
- A. SWAT
 - B. FTP
 - C. TCP
 - D. None of the above
9. SWAT service will run as _____ service.
- A. FTP
 - B. TCP
 - C. xinted
 - D. None of the above
10. Which of these utility is used to create a directory?
- A. crtdir
 - B. mkdir
 - C. dircreate
 - D. None of the above
11. Because of the SAMBA server, it is possible to
- A. Share files between computers
 - B. Share printers between computers
 - C. Both of the above
 - D. None of the above
12. SAMBA server on UNIX is an implementation of
- A. Windows SMB
 - B. CIFS protocol
 - C. Both of the above
 - D. None of the above
13. How can we start the RPM GUI tool?
- A. Main Menu | System Settings | Add/Remove Applications
 - B. \$ redhat-config-packages
 - C. By using either of the way mentioned above
 - D. Something other than this
14. Which service needs to be started for SAMBA server through service configuration?
- A. smbd
 - B. nmbd

- C. Both of the above mentioned
 - D. None of the above
15. By using which way, we can start the SAMBA service?
- A. Main Menu | System Settings | Server Settings | Services
 - B. `$ redhat-config-services`
 - C. By using either of the way mentioned above
 - D. None of the above

Answers for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. C | 2. B | 3. B | 4. C | 5. D |
| 6. A | 7. B | 8. A | 9. C | 10. B |
| 11. C | 12. C | 13. C | 14. C | 15. C |

Review Questions

1. What is a SAMBA server? What are the different ways to start RPM GUI tool for installation of SAMBA server?
2. What are the standard packages for SAMBA servers? How can we install the SAMBA server in the Linux system?
3. How can we start and stop the SAMBA service?
4. What are the different configuration files and utilities of SAMBA server? Explain.
5. What is SWAT? How can we install SWAT?
6. How can we start the SWAT service? How to use the SWAT service for first time?
7. How can we create and configure a SAMBA share?



Further Readings

Sandip Bhattacharya, Pancrazio De Mauro, Shishir Gundavaram, Mark Mamone, Kalip Sharma, Deepak Thomas and Simon Whiting, Beginning Red Hat Linux 9, Wiley Publishing, Inc.



Web Links

<https://linux.die.net/man/8/swat>